



2006
HIGHER SCHOOL CERTIFICATE
EXAMINATION

Biology

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- Draw diagrams using pencil
- Board-approved calculators may be used
- Write your Centre Number and Student Number at the top of pages 9, 11, 15 and 17

Total marks – 100

Section I Pages 2–21

75 marks

This section has two parts, Part A and Part B

Part A – 15 marks

- Attempt Questions 1–15
- Allow about 30 minutes for this part

Part B – 60 marks

- Attempt Questions 16–28
- Allow about 1 hour and 45 minutes for this part

Section II Pages 25–31

25 marks

- Attempt ONE question from Questions 29–33
- Allow about 45 minutes for this section

Section I

75 marks

Part A – 15 marks

Attempt Questions 1–15

Allow about 30 minutes for this part

Use the multiple-choice answer sheet.

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
 A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word **correct** and drawing an arrow as follows.

A ☒ B ☒ C ☐ D ☐
 correct ↖

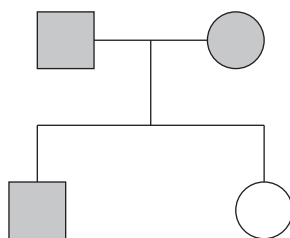
- 1** Which water treatment best reduces the risk of infection by killing pathogens in drinking water?
- (A) Chlorination
 - (B) Filtration
 - (C) Fluoridation
 - (D) Precipitation
- 2** What are the components of DNA?
- (A) Sugars, bases, proteins
 - (B) Sugars, phosphates, bases
 - (C) Phosphates, bases, polypeptides
 - (D) Phosphates, proteins, polypeptides
- 3** Which scientist carried out experiments that led to the development of an understanding of sex linkage?
- (A) Boveri
 - (B) Mendel
 - (C) Morgan
 - (D) Sutton
- 4** Evolutionary relationships between vertebrates can be determined by comparing the amino acid sequence of human haemoglobin with the haemoglobin of other vertebrates.
- Which area of study collects this type of evidence to support the theory of evolution?
- (A) Biochemistry
 - (B) Biogeography
 - (C) Comparative anatomy
 - (D) Comparative embryology

- 5 A student completed a genetics exercise by preparing the Punnett square drawn below. T represents a dominant allele and t represents a recessive allele.

		Parent 1 gametes	
		TT	Tt
Parent 2 gametes	Tt	Tt	tt
	tt		

What were the likely genotypes of the parents?

- (A) Parent 1 was homozygous, Parent 2 was heterozygous.
 (B) Parent 1 was heterozygous, Parent 2 was homozygous.
 (C) Both parents were heterozygous.
 (D) Both parents were homozygous.
- 6 A researcher prepared a pedigree (family tree) to trace a genetic disorder in a family.



Key

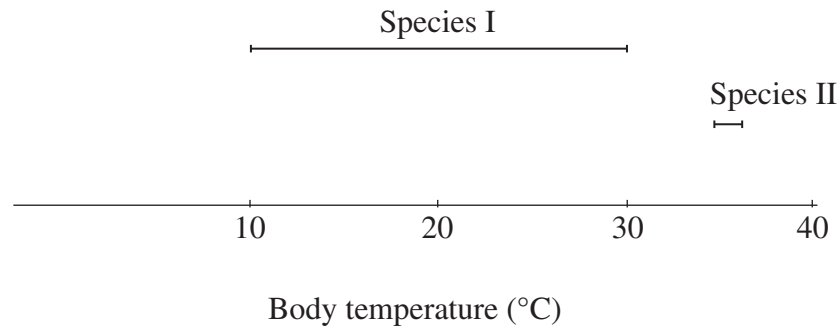
- Male has the genetic disorder
 ● Female has the genetic disorder
 ○ Female does not have the genetic disorder

How could the gene that causes the condition best be described?

- (A) Co-dominant
 (B) Dominant
 (C) Recessive
 (D) Sex-linked
- 7 Which alternative correctly identifies the tissues that transport carbohydrates?

	<i>Plant</i>	<i>Animal</i>
(A)	xylem	lymph
(B)	xylem	blood
(C)	phloem	lymph
(D)	phloem	blood

- 8 The ranges of body temperatures of two desert animal species are illustrated.



What is the best description to account for the range of body temperatures in Species I and Species II?

- (A) Species I is ectothermic; Species II is ectothermic.
 - (B) Species I is ectothermic; Species II is endothermic.
 - (C) Species I is endothermic; Species II is ectothermic.
 - (D) Species I is endothermic; Species II is endothermic.
- 9 What is the role of aldosterone in kidney function?
- (A) Aldosterone acts on the walls of the nephron tubules, changing the permeability of the tubules to react to changes in blood water concentration.
 - (B) Aldosterone is able to detect changes in blood pressure, and causes changes in water reabsorption to return blood pressure to normal.
 - (C) When blood pressure increases, aldosterone is released to increase sodium ion reabsorption rates, which causes less water to be reabsorbed, decreasing blood pressure.
 - (D) When blood pressure decreases, aldosterone is released to increase sodium ion reabsorption rates, which causes more water to be reabsorbed, increasing blood pressure.

- 10** An investigation was carried out where the composition of mammalian blood was compared before and after it had passed through different organs.

Data from one organ showed an increase in both carbon dioxide and digestion product levels, but a decrease in oxygen levels after passing through the organ.

Which organ would produce this result?

- (A) Gall bladder
 - (B) Kidney
 - (C) Pancreas
 - (D) Small intestine
- 11** The movement of materials through a plant occurs by translocation and the transpiration stream.

What is the main difference between these mechanisms?

- (A) Translocation occurs in endotherms, while the transpiration stream occurs in ectotherms.
 - (B) Translocation occurs in xylem tissue, while the transpiration stream occurs in phloem tissue.
 - (C) Translocation involves active transport, while the transpiration stream involves passive transport.
 - (D) Translocation transports respiration products, while the transpiration stream transports photosynthetic products.
- 12** Which two processes are important for the maintenance of health?
- (A) Mitosis and meiosis
 - (B) Mutation and mitosis
 - (C) Mitosis and cell differentiation
 - (D) Meiosis and cell differentiation

- 13** Allergies are the result of an immune response.

What triggers this response?

- (A) Antigens
- (B) Antidotes
- (C) Antibodies
- (D) Antihistamines

- 14** Organ transplant patients are given drugs to minimise the rejection of transplanted organs.

How do these drugs work?

- (A) They inhibit the production of enzymes by the transplanted organ that leads to its rejection.
- (B) They suppress the immune response which recognises 'foreign' molecules on the transplanted organs.
- (C) They act as antibiotics so that bacteria are killed before causing infections in transplanted organs.
- (D) They promote the repair of the blood vessel connections between the transplanted organ and the host body.

- 15** People who lived in the United Kingdom during the 1980s and 1990s are not accepted as blood donors in Australia because they might have eaten beef infected with prions.

Why is this precaution being taken?

- (A) Donated blood might contain prion toxins causing blood poisoning.
- (B) Donated blood might contain prions capable of causing viral infections.
- (C) Donated blood might contain prion DNA that results in prion replication in brain cells.
- (D) Donated blood might contain prions capable of altering protein structure in the brain cells.

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Centre Number

Section I (continued)

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Student Number

Part B – 60 marks

Attempt Questions 16–28

Allow about 1 hour and 45 minutes for this part

Answer the questions in the spaces provided.

Marks

Question 16 (3 marks)

(a) What is *haemoglobin*?

1

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(b) Describe ONE adaptive advantage of haemoglobin.

2

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Question 17 (3 marks)

Explain why it is necessary for organisms to remove metabolic waste products.

3

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Centre Number

Section I – Part B (continued)

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Student Number

Marks

Question 18 (7 marks)

- (a) Name ONE adaptation in an Australian terrestrial plant that assists in minimising water loss. **1**

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- (b) Explain why it is important for plant cells to control water loss. **2**

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- (c) Plant breeders have developed a new variety of terrestrial plant which has one structure that appears to assist in water conservation in hot, dry environments. **4**

Design a first-hand investigation the plant breeder could use to determine if this structure assists in water conservation.

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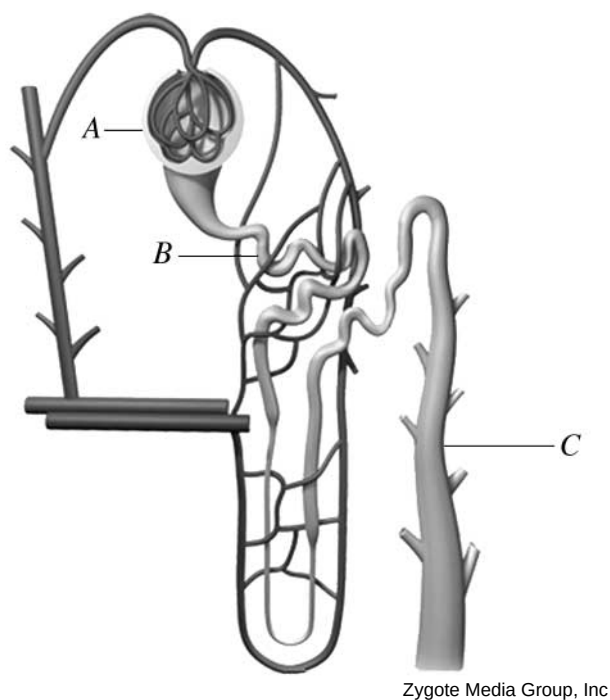
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Question 19 (5 marks)

The diagram shows a representation of a mammalian nephron.



- (a) Filtration happens at *A*. 1

Name TWO blood components that remain in the blood vessel after filtration.

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- (b) Most reabsorption happens at *B*. 2

Explain why BOTH passive and active transport are needed for this process to occur.

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- (c) ADH (anti-diuretic hormone) acts at *C*. 2

Outline the role of ADH in regulating water balance.

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Question 20 (3 marks)

A student bred a pair of goldfish and noticed that the offspring showed minor differences in appearance. **3**

Explain TWO processes that could result in these differences in appearance.

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Question 21 (4 marks)

(a) Summarise the evidence that led to Beadle and Tatum's 'one gene – one protein' hypothesis. **2**

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(b) Explain why the 'one gene – one protein' hypothesis was changed to the 'one gene – one polypeptide' hypothesis. **2**

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Centre Number

Section I – Part B (continued)

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Student Number

Marks**Question 22** (8 marks)

- (a) Identify a transgenic species, and state its use. **2**

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- (b) Outline ONE process used to produce a transgenic species. **2**

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- (c) Discuss ONE ethical issue arising from the use of transgenic species. **4**

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Question 23 (4 marks)

In 1926, T H Muller experimented with fruit flies (*Drosophila* sp.) by exposing them to X-rays. He found that their offspring showed new phenotypes not observed in the wild population.

4

Explain how the results of these experiments can provide support for Darwin's theory of evolution by natural selection.

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Centre Number

Section I – Part B (continued)

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Student Number

Marks

Question 24 (2 marks)

Black spot is an infectious apple disease that occurs in New Zealand and makes apples unsuitable for sale. **2**

Describe a method to prevent the spread of the disease into Australia.

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Question 25 (3 marks)

Using an example, describe how a disease can be caused by an imbalance of microflora in humans. **3**

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Question 26 (8 marks)**8**

Awaiting Copyright Clearance

Warren and Marshall's investigations included:

- using a microscope to look at prepared slides of ulcerated stomach tissues;
- using a flexible endoscope to look into the stomach of patients with stomach ulcers and gastritis (localised or general inflammation of the stomach);
- using staining techniques to determine the possible presence of bacteria in stomach tissue.

In addition, Warren checked that Marshall's stomach contained no *Helicobacter pylori*. Marshall then swallowed a dose of the bacteria, triggering symptoms of gastritis.

Assess the procedures that Warren and Marshall used to identify and confirm their conclusion about the pathogen that caused stomach ulcers and gastritis.

Question 26 continues on page 19

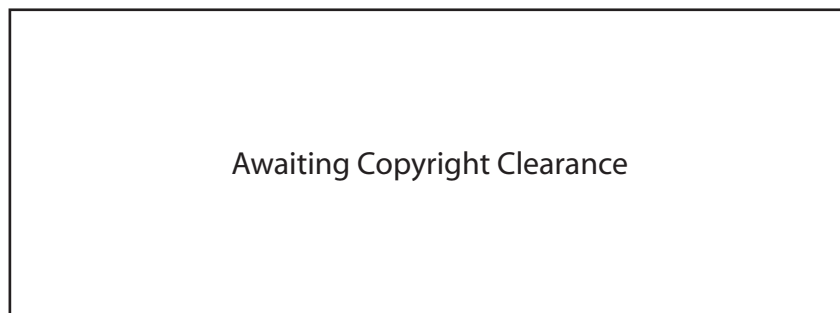
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End of Question 26

Question 27 (4 marks)

Lactose intolerance is a non-infectious disease found in some humans who cannot digest milk. The causes of non-infectious disease are grouped into three categories. An epidemiological study was undertaken to investigate the cause of lactose intolerance. The study found variation in the occurrence of lactose intolerance in human populations from different parts of the world. Some of the data are shown in the table.

4



Using information from the table, outline how the study could be continued to determine the cause of the disease.

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Question 28 (6 marks)

- (a) A gardener noticed a red and swollen area on his arm that had received a deep scratch from a thorn on a plant. **1**

Identify the most likely human defence adaptation that caused these symptoms.

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- (b) After a number of days, the gardener's arm remained red and swollen, so he visited his doctor who prescribed an antibiotic to treat the infection. **1**

Why did the doctor prescribe an antibiotic to treat the infection?

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- (c) The gardener should have taken the antibiotic for ten days but stopped after five days because the arm was no longer red or swollen. **4**

Explain how this action by the gardener might lead to antibiotic resistance.

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Biology

Section II

25 marks

Attempt ONE question from Questions 29–33

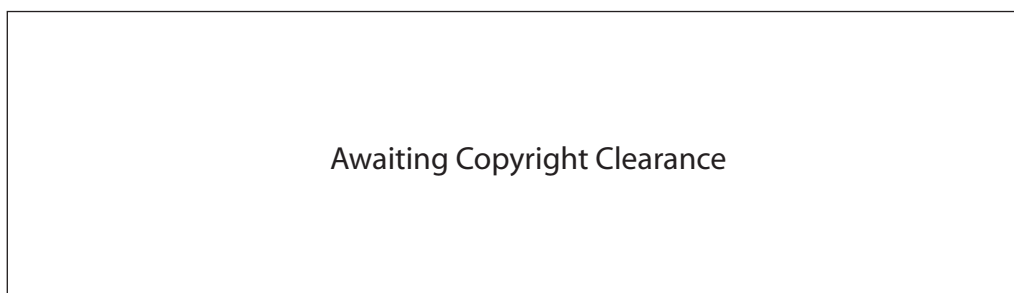
Allow about 45 minutes for this section

Answer the question in a writing booklet. Extra writing booklets are available.

	Pages
Question 29 Communication	26
Question 30 Biotechnology	27
Question 31 Genetics: The Code Broken?	28–29
Question 32 The Human Story	30
Question 33 Biochemistry	31

Question 29 — Communication (25 marks)

- | | | |
|-----|--|----------|
| (a) | (i) Outline three differences between rods and cones in the retina. | 3 |
| | (ii) Explain their role in human colour blindness. | 2 |
| (b) | (i) Describe the procedure used, and outline the results obtained, when you modelled the process of accommodation. | 5 |
| | (ii) State how using this model increases understanding of accommodation. | 1 |
| (c) | Discuss how advances in hearing and vision technology have assisted people with difficulties in communication. | 7 |
| (d) | A student located the following information about brain damage. | |



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|-------|---|----------|
| (i) | Identify the organ that detects light stimuli in humans. | 1 |
| (ii) | Outline ONE piece of evidence that half of the brain is functioning correctly when a human is affected by <i>visual neglect</i> . | 2 |
| (iii) | Explain the role of the brain in the coordination of animal behaviour. | 4 |

Question 30 — Biotechnology (25 marks)

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|-----|---|----------|
| (a) | (i) Outline ONE use of biotechnology in an early society. | 3 |
| | (ii) Explain why the use you outlined in part (a) (i) may be considered an early biotechnology. | 2 |
| (b) | (i) Describe the procedure used, and outline the results obtained, when you demonstrated the fermentation processes in bread or alcohol production. | 5 |
| | (ii) State how using this demonstration increases understanding of fermentation processes. | 1 |
| (c) | Discuss the impact of the development of recombinant DNA technology on applications of biotechnology. | 7 |
| (d) | A student researching the impact of ethical issues on decision-making processes in biotechnology located the following information. | |

A plant commonly used for animal food was deficient in the amino acid, methionine. The nutritional value of the plant was improved by adding a gene from a different plant species rich in methionine.

However, the gene came from a plant to which some people showed an allergic reaction when exposed to the plant's products. When these people were exposed to the transgenic plant they also showed an allergic reaction.

The transgenic plant was going to be marketed only for animal feed, but the difficulty of keeping animal feed separate from human food during harvesting, transport, and storage became a consideration. It was decided not to ask for approval to market the transgenic plant. It was never approved by the government and was never grown commercially or sold in stores.

- | | | |
|-------|--|----------|
| (i) | Identify ONE potential benefit of using the transgenic plant. | 1 |
| (ii) | Outline ONE piece of evidence supporting the views of a group opposed to using the transgenic plant. | 2 |
| (iii) | Evaluate decisions relating to the commercial use of cloning or genetically modified organisms. | 4 |

Adapted from web site 'Transgenic Crops: An Introduction and Resource Guide' By Department of Soil & Crop Sciences, Colorado State University

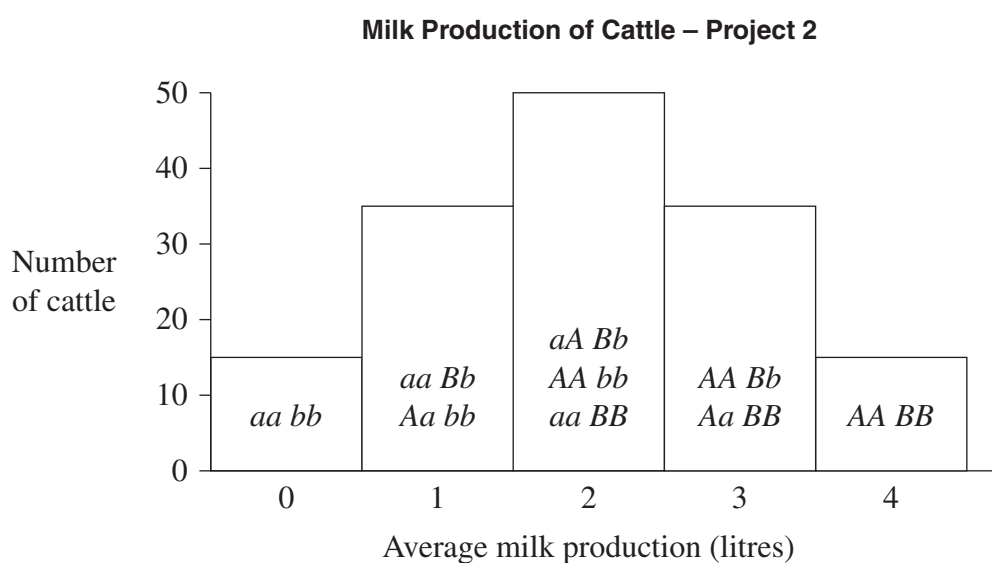
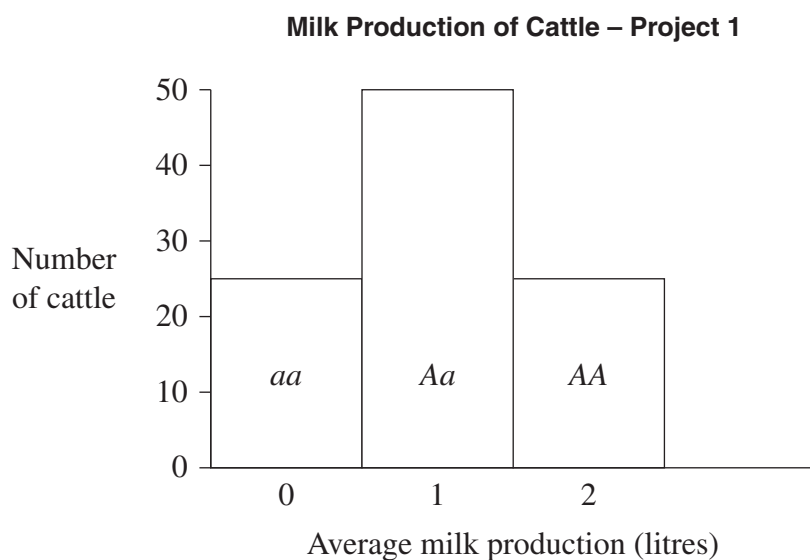
Question 31 — Genetics: The Code Broken? (25 marks)

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|-----|------|--|----------|
| (a) | (i) | Outline THREE differences between somatic and gametic cells. | 3 |
| | (ii) | Explain ONE reason for the differences as outlined in part (a) (i) in terms of their importance in inheritance. | 2 |
| (b) | (i) | Describe the procedure used, and outline the findings obtained, when you processed information on current understanding of gene expression. | 5 |
| | (ii) | State how the findings increased an understanding of gene expression. | 1 |
| (c) | | Discuss how major advances in our knowledge of genetics have changed our understanding of the way genes direct the structure, function and development of an organism. | 7 |

Question 31 continues on page 29

Question 31 (continued)

- (d) A scientist carried out research projects for a farmers' organisation to identify the link between cattle genotype and milk production. The graphs show the average milk production of cattle with different genotypes.

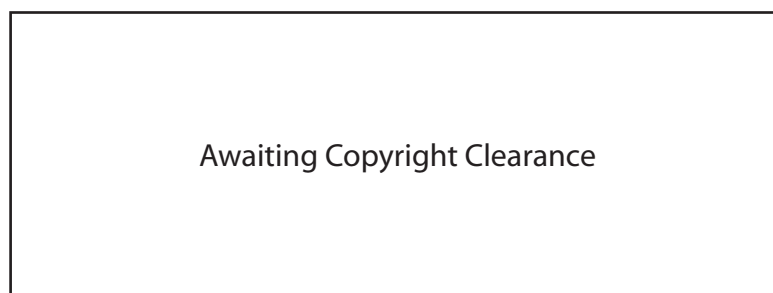


- (i) Identify the cattle genotype that produces the greatest amount of milk. **1**
- (ii) Outline the effect on milk production of cattle having one or more dominant allele. **2**
- (iii) Explain the effect of polygenic inheritance on variability within a trait. **4**

End of Question 31

Question 32 — The Human Story (25 marks)

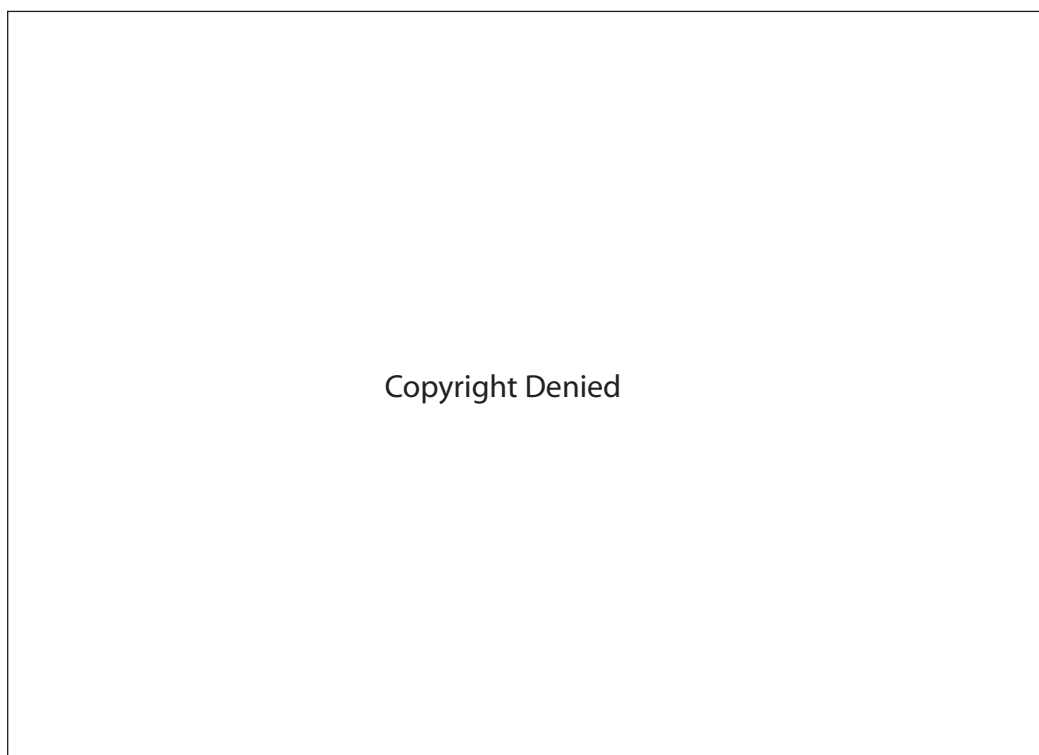
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|-----|--|----------|
| (a) | (i) Name TWO examples of cultural change that occurred as humans developed into efficient hunters in organised cooperative groups. | 2 |
| | (ii) Explain how ONE of the cultural changes named in part (a) (i) has been a feature of human evolution. | 3 |
| (b) | (i) Describe the procedure used, and outline the findings obtained, when you investigated the maternal inheritance of mitochondrial DNA. | 5 |
| | (ii) State the importance of the study of the maternal inheritance of mitochondrial DNA in tracing human evolution. | 1 |
| (c) | Discuss how advances in technology have changed our understanding of the evolutionary relationships between humans and other primates. | 7 |
| (d) | The chart shows the dispersal of the genus <i>Homo</i> . | |



- | | | |
|-------|---|----------|
| (i) | Identify ONE species of <i>Homo</i> thought to be living one million years ago. | 1 |
| (ii) | Outline ONE piece of information to support the hypothesis that <i>Homo floresiensis</i> evolved from <i>Homo erectus</i> . | 2 |
| (iii) | Based on the information in the chart, assess how the discovery of <i>Homo floresiensis</i> , living at the same time as <i>Homo sapiens</i> , could be used to support EITHER the ‘Out of Africa’ model OR the ‘Theory of Regional Continuity’ as an explanation for the pattern of human migration and evolution. | 4 |

Question 33 — Biochemistry (25 marks)

- | | | |
|-----|---|----------|
| (a) | (i) State TWO reasons for studying photosynthesis. | 2 |
| | (ii) Outline the contribution of ONE 17th or 18th century scientist in developing our understanding of photosynthesis. | 3 |
| (b) | (i) Describe the procedure used, and outline the results obtained, when you extracted the mixture of pigments from leaves and examined their absorption spectrum. | 5 |
| | (ii) State how the results have increased an understanding of the role of pigments in photosynthesis. | 1 |
| (c) | Discuss the impact of improvements in microscopy on our understanding of cell structures involved in photosynthesis. | 7 |
| (d) | The diagram shows the Calvin Cycle. | |



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|-------|---|----------|
| (i) | Identify molecule X. | 1 |
| (ii) | State the significance of ATP at (Z) and (Y) in the Calvin Cycle. | 2 |
| (iii) | Assess how knowledge and understanding of the Calvin Cycle has increased with advances in technology. | 4 |

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2006 HSC Notes from the Marking Centre Biology

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2006 HSC NOTES FROM THE MARKING CENTRE

BIOLOGY

Introduction

This document has been produced for the teachers and candidates of the Stage 6 course in Biology. It provides comments with regard to responses to some questions in the 2006 Higher School Certificate Examination, indicating the quality of candidate responses and highlighting the relative strengths and weaknesses of the candidature in each section.

This document should be read along with the relevant syllabus, the 2006 Higher School Certificate examination, the marking guidelines and other support documents which have been developed by the Board of Studies to assist in the teaching and learning of Biology.

General Comments

In 2006, approximately 14,000 candidates attempted the Biology examination. The most popular Options were Communication (59%), and Genetics: The Code Broken? (20%).

Teachers and candidates should be aware that examiners may write questions that address the syllabus outcomes in a manner that requires candidates to respond by integrating their knowledge, understanding and skills developed through studying the course, including the Prescribed Focus Areas. This reflects the fact that the knowledge, understanding and skills developed through the study of discrete sections should accumulate to a more comprehensive understanding than may be described in each section separately. It is important to understand that the Preliminary HSC course is assumed knowledge for the HSC course.

In 2006, at least one question in Section I Part B and one part of the Section II option questions focused on the mandatory skills content in Module 9.1. Candidates who had actively planned and performed practical experiences clearly demonstrated a deeper knowledge and understanding of the content described in this module.

Overall, the level of understanding of Biology concepts indicated by responses was appropriate for most HSC candidates. Candidates need to be reminded that the answer space provided and the marks allocated are guides to the maximum length of response required. Similarly, the key word used in the question gives an indication of the depth of the required response. Candidates should use examination time to analyse the question and plan responses carefully, working within that framework to produce a clear and concise response. This may include the use of dot points, diagrams and/or tables, and avoid internal contradictions. This is particularly important in holistic questions which need to be logical and well structured. There was evidence that some candidates had a very poor knowledge of basic definitions specific to terminology associated with the course.

Better responses indicate that candidates are following the instructions provided on the examination paper. In these responses, candidates:

- do not repeat the question as part of the response
- look at the structure of the whole question and note that in some questions the parts follow from each other (ie responses in part (a) lead to the required response in part (b) etc).

In Section II, the option question is divided into a number of parts. Candidates should clearly label each part of the question when writing in their answer booklets. In part (c) of the 2006 option questions, the best responses presented ideas coherently and included the correct use of scientific principles and ideas. Many candidates wrote a lot of information that was not relevant to the question. Some responses showed evidence of rote learning of an anticipated answer based on a single source. These responses did not address the syllabus content and/or outcomes being assessed and hence did not score full marks. Candidates are required to attempt one question only in Section II, but some candidates responded to more than one option question. Candidates are strongly advised to answer the option they have studied in class.

Section I – Core

Part A – Multiple-choice questions

Question	Correct Response
1	A
2	B
3	C
4	A
5	C
6	B
7	D
8	B

Question	Correct Response
9	D
10	D
11	C
12	C
13	A
14	B
15	D

Part B

Specific Comments

Question 16

- (a) The majority of candidates were able to state a function, structure or location of haemoglobin.
- (b) Better responses demonstrated an understanding of the advantage of haemoglobin. Weaker responses showed knowledge of haemoglobin's structure and/or function without addressing adaptive advantage.

Question 17

The majority of candidates demonstrated a sound understanding of the need for organisms to remove metabolic waste products.

Question 18

- (a) Most candidates were able to name an adaptation.
- (b) Better responses were able to link the maintenance of water balance with a correct physiological reason. Weaker responses did not relate cause and effect.

- (c) Better responses outlined the features of good experimental design, including aspects such as control of variables, repetition and data collection.

Question 21

- (a) While better responses demonstrated an understanding of the ‘one gene – one protein’ hypothesis, a significant number of responses demonstrated a lack of knowledge and/or understanding of Beadle and Tatum's experiment.
- (b) Better responses demonstrated an understanding that hypotheses are modified to account for new knowledge. Weaker responses showed confusion about the relationship between genes, proteins and polypeptides.

Question 22

- (a) Better responses named a transgenic species and its use. Weaker responses gave a general definition but no specific example.
- (b) Better responses correctly provided the main features of one process used to produce a transgenic species.
- (c) Better responses provided points for and/or against an identified ethical issue arising from the use of a transgenic species.

Question 23

In the better responses, candidates were able to relate Darwin's ideas about the nature and sources of variation to the experiment in which new alleles/varieties by mutagenic radiation were formed.

Question 24

The majority of candidates were able to describe an appropriate method to prevent the spread of the disease “Black spot”.

Question 25

Better responses described for a named disease the cause and general effects of an imbalance and clearly distinguished between the effects on microflora and the symptoms of the disease.

Question 26

In the better responses, candidates were able to make a judgement about the procedures using the supporting information provided in the question and referring to the use of different technologies, a variety of methods and one issue. Weaker responses often outlined or described a different approach but failed to make a judgement about the procedures.

Question 27

In the better responses, candidates indicated the main features of an appropriate investigation, chose appropriate groups to study, and made reference to both the table and each category.

Question 28

- (a) Most candidates correctly identified inflammation.
- (b) Most candidates identified bacteria as the source of infection.
- (c) In the better responses, candidates linked the failure to complete the course of antibiotics with how natural selection leads to antibiotic resistance. Weaker responses confused Darwinian and Lamarckian concepts of evolution.

Section II - Options

Question 29 – Communication

- (a) (i) Better responses provided corresponding features for both rods and cones rather than features for one with implied differences. A table was often used to effectively and explicitly present the differences.
- (ii) Better responses clearly showed the cause and effect relationship between the lack of pigments in the cones and colour-blindness.
- (b) (i) Better responses provided the main features of an appropriate procedure with risk assessments and detailed information about the relationship of the parts of their model to the parts of the eye. Many responses included accurate labelled diagrams and demonstrated an understanding of the concept of modelling and the process of accommodation.
- (ii) In better responses, candidates were able to clearly identify how their model increased their understanding of accommodation.
- (c) Better responses identified a range of hearing and vision technologies and provided detailed descriptions of the benefits and/or limitations of these technologies. Weaker responses were excessively long, showing little evidence of planning or structure.
- (d) (i) Better responses stated that the eye is the organ that detects light stimuli in humans. A significant number of candidates called it a tissue rather than an organ.
- (ii) Most candidates were able to identify one piece of evidence from the stimulus material. The better responses recognised that some brain function was required to process the sensory input.
- (iii) Most candidates were able to describe the role of the brain in coordination. In the better responses, candidates provided a clear link between loss of brain function and impaired coordination.

Question 30 – Biotechnology

- (a) (i) In the better responses, candidates identified and sketched in general terms the use of biotechnology in a named early society.
- (ii) Better responses clearly linked the use of biotechnology in an early society with the use of organisms to make or modify a product for human benefit.
- (b) (i) The better responses provided the main features of an appropriate procedure with risk assessment and an outline of the results obtained. Weaker responses indicated some confusion between the processes of fermentation and the baking of bread.
- (ii) Better responses clearly identified how the demonstration increased understanding of the fermentation process. However, this question proved challenging for many candidates.

- (c) Better responses identified the issues related to the development of recombinant DNA technology and provided detailed descriptions of the benefits and/or limitations of the applications of this technology. Weaker responses were excessively long, showing little evidence of planning or structure.
- (d) (i) Most candidates were able to use the stimulus material to determine that the transgenic plant had improved nutritional value.
- (ii) In most responses, candidates were able to provide the main feature of the supporting evidence.
- (iii) Most responses made judgements about the commercial use of cloning or genetically modified organisms. Better responses provided evidence to support the judgement.

Question 31 – Genetics: The Code Broken?

- (a) (i) In the better responses, candidates provided corresponding features for both somatic and gametic cells rather than features for one with implied differences. A table was often used to effectively and explicitly present the differences.
- (b) (i) Better responses provided main features of an appropriate procedure used and the findings obtained about the current understanding of gene expression. Many candidates demonstrated a good understanding of the role of homeotic genes, switching genes on and off, gene cascades and limb development.
- (ii) Better responses clearly identified how the findings increased their understanding of the gene expression.
- (c) Better responses described major advances in our knowledge of genetics that have changed our understanding of the way genes direct structure, function and development and described benefits and/or limitations of our changed understanding. Weaker responses were excessively long, showing little evidence of planning or structure.
- (d) (i) In the better responses, candidates identified the correct genotype that produced the greatest amount of milk as AABB.
- (ii) In most responses, candidates provided, in general terms, the effect of the dominant allele on milk production.
- (iii) Better responses clearly linked polygenic inheritance with greater variability within a trait.

Question 32 – The Human Story

- (a) (ii) In the better responses, candidates linked the named cultural change to an aspect of human evolution.
- (b) (i) Better responses provided the main features of an appropriate procedure and used the findings obtained about the maternal inheritance of mitochondrial DNA. Weaker responses described an irrelevant first-hand investigation.

- (ii) Most candidates clearly identified the importance of the study of maternal inheritance of mitochondrial DNA.
- (c) Better responses identified a range of technologies such as karyotype analysis, DNA-DNA hybridisation and mitochondrial DNA sequencing, and provided detailed descriptions of the benefits and/or limitations of these technologies in changing our understanding of the evolutionary relationships between humans and other primates. Weaker responses were excessively long, showing little evidence of planning or structure.
- (d) (i) Most candidates were able to use the stimulus material to correctly recognise and name *Homo erectus*.
- (ii) Better responses used the stimulus material to support the hypothesis.
- (iii) In the better responses, candidates referred to the stimulus material to link and provide a judgement as to how this discovery supported the selected model.

Question 33 – Biochemistry

- (a) (ii) Better responses identified the scientist and their contribution to our understanding of photosynthesis.
- (b) (i) Better responses provided the main features of an appropriate procedure for examining the absorption spectrum with risk assessments and an outline of the results obtained.
- (ii) Few candidates were able to show how the results increased their understanding of the role of pigments in photosynthesis.
- (c) Better responses identified a range of improvements in microscopy and provided detailed descriptions of the benefits and/or limitations of these improvements on increasing our understanding of cell structures involved in photosynthesis. Weaker responses were excessively long, showing little evidence of planning or structure.
- (d) (i) Better responses correctly named X as phosphoglycerate.
- (ii) Better responses identified the significance of ATP providing energy at both points Z and Y.
- (iii) Better responses described, linked and made a judgement as to how a specific advance in technology increased understanding of the Calvin Cycle.

Biology

2006 HSC Examination Mapping Grid

Question	Marks	Content	Syllabus outcomes
Section I Part A			
1	1	9.4.2.3.2	H8
2	1	9.3.3.2.3	H9
3	1	9.3.3.2.7	H9
4	1	9.3.1.2.2	H10
5	1	9.3.2.2.4, 9.3.2.3.2, 14.1 (a)	H14
6	1	9.3.2.3.1, 12.3 (c), 14.1 (c)	H12, H14
7	1	9.2.2.2.1, 9.2.2.2.6	H6
8	1	9.2.1.2.8, 12.3 (c), 14.1 (a)	H12, H14
9	1	9.2.3.2.7	H6
10	1	9.2.2.2.4, 14.3 (c)	H14
11	1	9.2.2.2.6	H6
12	1	9.4.1.2.2	H6
13	1	9.4.4.2.2	H6
14	1	9.4.5.2.4, 9.4.4.2.3	H3, H6
15	1	9.4.3.2.2	H3, H6
Section I Part B			
16 (a)	1	9.2.2.2.2	H6
16 (b)	2	9.2.2.2.2	H6, H10
17	3	9.2.3.2.2	H6
18 (a)	1	9.2.3.2.9	H6
18 (b)	2	9.2.3.2.1	H6
18 (c)	4	9.2.3.3.7, 11.2 (c), 14.3 (a)	H11, H14
19 (a)	1	9.2.3.2.6	H6

Question	Marks	Content	Syllabus outcomes
19 (b)	2	9.2.3.2.5	H6
19 (c)	2	9.2.3.2.7	H6
20	3	9.3.3.2.5, 9.3.3.3.3	H9
21 (a)	2	9.3.4.3.2	H6, H9
21 (b)	2	9.3.4.3.2	H3, H6, H9
22 (a)	2	9.3.5.3.2	H4, H7
22 (b)	2	9.3.5.2.2	H3
22 (c)	4	9.3.5.3.2	H4, H10
23	4	9.3.4.2.4, 9.3.4.2.6, 9.3.4.2.5	H9, H10
24	2	9.4.7.2.1	H3, H8
25	3	9.4.4.3.1	H8
26	8	9.1, 9.4.3.1	H11, H12, H13
27	4	9.4.6.1, 9.4.6.2.2, 9.4.6.3.2, 9.1, 11.1 (e), 14.1 (c)	H6, H11, H14
28 (a)	1	9.4.4.2.4	H6
28 (b)	1	9.4.3.2.3	H6
28 (c)	4	9.3.4.3.4, 9.4.3.3.4	H10
Section II			
Question 29 — Communication			
29 (a) (i)	3	9.5.4.2.2	H6
29 (a) (ii)	2	9.5.4.2.5	H6
29 (b) (i)	5	9.5.3.3.1, 11.2 (d), 11.3 (a,b,c)	H11
29 (b) (ii)	1	9.5.3.3.1, 14.1 (b), 14.3 (c)	H14
29 (c)	7	9.5.3, 9.5.6, 14.3 (b)	H3, H14
29 (d) (i)	1	9.5.2.2.1	H6
29 (d) (ii)	2	9.5.7.2.4, 14.1 (b)	H14
29 (d) (iii)	4	9.5.7.2.5	H6

Question	Marks	Content	Syllabus outcomes
Section II			
Question 30 — Biotechnology			
30 (a) (i)	3	9.6.1.2.1	H8
30 (a) (ii)	2	9.6.1.2.2	H8
30 (b) (i)	5	9.6.2.3.1, 11.3 (a,b,c), 11.2 (d)	H11
30 (b) (ii)	1	9.6.2.3.1, 14.1 (b), 14.3 (c)	H14
30 (c)	7	9.6.5.1, 9.6.6.1, 14.3 (b)	H1, H14
30 (d) (i)	1	9.6.7.3.1, 12.3 (c)	H12
30 (d) (ii)	2	9.6.7.2.2, 9.6.7.3, 14.1 (b)	H14
30 (d) (iii)	4	9.6.7	H7
Section II			
Question 31 — Genetics: The Code Broken?			
31 (a) (i)	3	9.7.3.2.1	H9
31 (a) (ii)	2	9.7.3.2.1, 9.7.3.2.2, 9.7.3.3.1	H9
31 (b) (i)	5	9.7.1.3.2, 14.1 (c), 11.1 (b), 12.3 (a, d) 12.4 (d, e)	H11, H12, H14
31 (b) (ii)	1	9.7.1.3.2, 14.3 (c)	H14
31 (c)	7	9.7, 9.7.8, 14.3 (b)	H1, H14
31 (d) (i)	1	9.7.2.2.1, 9.7.2.3.2, 12.3 (c), 14.1 (a)	H12, H14
31 (d) (ii)	2	9.7.2.2.1, 9.7.2.3.2, 12.3 (c)	H12
31 (d) (iii)	4	9.7.2.2.3, 9.7.2.3.2, 14.1 (a)	H9, H14
Section II			
Question 32 — The Human Story			
32 (a) (i)	2	9.8.5.2.1	H10
32 (a) (ii)	3	9.8.5.2.2	H10
32 (b) (i)	5	9.8.2.3.3, 11.1 (b), 12.3 (a, b), 14.1 (c)	H10, H11, H12, H14
32 (b) (ii)	1	9.8.4.3.1, 14.1 (h), 14.3 (b)	H10, H14
32 (c)	7	9.8.1.3.3, 9.8.2, 14.3 (b)	H1, H14
32 (d) (i)	1	9.8.3, 12.3 (c)	H10, H12
32 (d) (ii)	2	9.8.3, 12.3 (c)	H10, H12

Question	Marks	Content	Syllabus outcomes
32 (d) (iii)	4	9.8.3.2.3, 14.1 (c)	H10, H14
Section II			
Question 33 — Biochemistry			
33 (a) (i)	2	9.9.1.2.1	H6
33 (a) (ii)	3	9.9.2.2.1	H1, H6
33 (b) (i)	5	9.9.3.3.4, 11.2 (d), 11.3 (a, b, c)	H11
33 (b) (ii)	1	9.9.3.3.4, 14.1 (b)	H14, H12
33 (c)	7	9.9.8, 9.9.3.3.2, 14.3 (b)	H1, H14
33 (d) (i)	1	9.9.6.2.3, 12.3 (c)	H6, H12
33 (d) (ii)	2	9.9.6.2.3, 9.9.6.3	H6
33 (d) (iii)	4	9.9.6, 14.1 (b), 14.1 (g)	H2, H14

2006 HSC Biology Marking Guidelines

Section I, Part B

Question 16 (a)

Outcomes assessed: H6

MARKING GUIDELINES

Criteria	Marks
• Any ONE structure, function or location	1

Question 16 (b)

Outcomes assessed: H6, H10

MARKING GUIDELINES

Criteria	Marks
• Describes ONE advantage	2
• Identify ONE advantage	1

Question 17

Outcomes assessed: H6

MARKING GUIDELINES

Criteria	Marks
• Links maintaining normal function of cells/tissues/organs/organisms with accumulation of waste products AND changes in chemical balance	3
• Links any two of maintaining normal function of cells/tissues/organs/organisms OR accumulation of wastes OR changes in chemical balance	2
• States ONE correct reason	1

Question 18 (a)

Outcomes assessed: H6

MARKING GUIDELINES

Criteria	Marks
• Names one CORRECT adaptation	1

Question 18 (b)

Outcomes assessed: H6

MARKING GUIDELINES

Criteria	Marks
• Links maintaining water balance with ONE correct physiological reason	2
• States ONE physiological reason	1

Question 18 (c)

Outcomes assessed: H11, H14

MARKING GUIDELINES

Criteria	Marks
• Outlines method including variables, repetition, controls and data to be collected about the structure	3–4
• Brief or incomplete description includes some elements of a first-hand investigation	1–2

Question 19 (a)

Outcomes assessed: H6

MARKING GUIDELINES

Criteria	Marks
• Correctly names TWO blood components	1

Question 19 (b)

Outcomes assessed: H6

MARKING GUIDELINES

Criteria	Marks
• Correctly explains why BOTH passive and active transport are needed	2
• Correctly explains why EITHER passive OR active transport are needed	1

Question 19 (c)

Outcomes assessed: H6

MARKING GUIDELINES

Criteria	Marks
• Correctly sequences a change in water balance with the release/inhibition of ADH with effect	2
• Incomplete sequence	1

Question 20

Outcomes assessed: H9

MARKING GUIDELINES

Criteria	Marks
• Correctly explains TWO processes (which can include genetic and/or environmental processes)	3
• Correctly explains ONE process	2
• Only identifies process(es)	1

Question 21 (a)

Outcomes assessed: H6, H9

MARKING GUIDELINES

Criteria	Marks
• Completely summary of evidence	2
• Incomplete summary of evidence	1

Question 21 (b)

Outcomes assessed: H3, H6, H9

MARKING GUIDELINES

Criteria	Marks
• Explains the need to modify the hypothesis to account for new knowledge	2
• Outlines appropriate new knowledge	1

Question 22 (a)

Outcomes assessed: H4, H7

MARKING GUIDELINES

Criteria	Marks
• Correctly identifies one transgenic species and states its use	2
• Correctly identifies one transgenic species OR • States a use for any transgenic species	1

Question 22 (b)

Outcomes assessed: H3

MARKING GUIDELINES

Criteria	Marks
• Correctly outlines one process	2
• Identifies one process	1

Question 22 (c)

Outcomes assessed: H4, H10

MARKING GUIDELINES

Criteria	Marks
• Names one ethical issue and discusses at least two points for and/or against OR • Names one ethical issue and identifies at least three points for and/or against	3–4
• Names one ethical issue and provides one point for or against	2
• Names one ethical issue OR • Provides one point for or against	1

Question 23

Outcomes assessed: H9, H10

MARKING GUIDELINES

Criteria	Marks
Links Darwin's understanding of the nature and source of variation in populations with the formation of new alleles/varieties by mutagenic radiation	3–4
- Identifies the aspect of Darwin's Theory of Evolution by Natural Selection by the experiment and identifies the cause of mutation OR - Outlines the source of variation as mutagenic radiation	2
- Identifies the aspect of Darwin's Theory of Evolution by Natural Selection supported by the experiment OR - Identifies the cause of mutation	1

Question 24

Outcomes assessed: H3, H8

MARKING GUIDELINES

Criteria	Marks
Describes appropriate method	2
Identifies appropriate method	1

Question 25

Outcomes assessed: H8

MARKING GUIDELINES

Criteria	Marks
Identifies an appropriate disease AND describes the cause AND general effects of an imbalance	3
- Identifies an appropriate disease - Describes a cause OR - Describes general effects of an imbalance	2
- Identifies an appropriate disease OR - Outlines a cause OR - Outlines general effects of an imbalance	1

Question 26

Outcomes assessed: H11, H12, H13

MARKING GUIDELINES

Criteria	Marks
Makes a judgement using supporting information including reference to the use of different technologies, a variety of methods and at least one issue	7–8
Describes approaches using supporting information including references to the use of different technologies, a variety of methods and at least one issue	5–6
Outlines approaches with reference to different technologies, a variety of methods, and one issue	3–4
- States TWO of the following: - identifies that a variety of technologies were used - identifies that a variety of methods were used - identifies one appropriate issue	2
- Identifies one appropriate issue OR - Identifies that a variety of technologies were used OR - Identifies that a variety of methods were used OR - Identifies a method and a technology used	1

Question 27

Outcomes assessed: H6, H11, H14

MARKING GUIDELINES

Criteria	Marks
• Outlines an appropriate investigation referring to each category AND referring to the table	4
• Outlines an appropriate investigation referring to TWO categories AND referring to the table	3
• Outlines an investigation referring to ONE category AND referring to the table OR • Outlines an investigation referring to each category without referring to the table	2
• Outlines an investigation referring to ONE category	1

Question 28 (a)

Outcomes assessed: H6

MARKING GUIDELINES

Criteria	Marks
• Identifies correct human defence adaptation	1

Question 28 (b)

Outcomes assessed: H6

MARKING GUIDELINES

Criteria	Marks
• Identifies bacteria as the source of infection	1

Question 28 (c)

Outcomes assessed: H10

MARKING GUIDELINES

Criteria	Marks
• Links failure to complete the course of antibiotics with how natural selection leads to antibiotic resistance	4
• Outlines how natural selection leads to antibiotic resistance	2–3
• Identifies that natural selection can lead to antibiotic resistance OR • Identifies failure to complete the course of antibiotics as selecting agent	1

Section II

Question 29 (a) (i)

Outcomes assessed: H6

MARKING GUIDELINES

Criteria	Marks
• Outlines THREE differences	3
• Outlines TWO differences	2
• Outlines ONE difference	1

Question 29 (a) (ii)

Outcomes assessed: H6

MARKING GUIDELINES

Criteria	Marks
• Links absence of at least one of the types of cones with colour blindness	2
• Identifies that cones are involved with colour vision OR • Identifies that rods are not involved	1

Question 29 (b) (i)

Outcomes assessed: H11

MARKING GUIDELINES

Criteria	Marks
• Appropriate procedure with relevant results, including detailed information about procedure AND result	5
• Appropriate procedure AND/OR relevant results	3–4
• Limited procedure AND/OR • Limited result	1–2

Question 29 (b) (ii)

Outcomes assessed: H14

MARKING GUIDELINES

Criteria	Marks
Any relevant statement	1

Question 29 (c)

Outcomes assessed: H3, H14

MARKING GUIDELINES

Criteria	Marks
- Describes how at least one vision and one hearing technology have assisted people - Discusses benefits and/or limitation of advances in both hearing and vision technology - Provides a response that demonstrates coherence and logical progression and includes correct use of scientific principles and ideas	7
- Describes how one vision and one hearing technology have assisted people - Describe the benefits or limitations of at least one technology in both vision and hearing	5–6
- Describes how two advances have assisted people - Identifies at least one benefit or limitation of each advance	3–4
- Outline how one advance in hearing or vision technology has assisted people OR - Identifies two advances in hearing or vision technology	2
Identifies one advance in hearing or vision technology	1

Question 29 (d) (i)

Outcomes assessed: H6

MARKING GUIDELINES

Criteria	Marks
Identifies the eye	1

Question 29 (d) (ii)

Outcomes assessed: H14

MARKING GUIDELINES

Criteria	Marks
• Outlines one piece of evidence	2
• Identifies one piece of evidence	1

Question 29 (d) (iii)

Outcomes assessed: H6

MARKING GUIDELINES

Criteria	Marks
• Links the description of the correct functioning of the brain with coordination of behaviour	4
• Links the description of one role of the brain with coordination	3
• Describes one role of the brain	2
• Outlines one role of the brain	1

Question 30 (a) (i)

Outcomes assessed: H8

MARKING GUIDELINES

Criteria	Marks
• Outlines TWO features of the use of biotechnology in an early society	3
• Outlines ONE feature of the use of biotechnology in an early society	2
• Identifies ONE use of biotechnology in an early society	1

Question 30 (a) (ii)

Outcomes assessed: H8

MARKING GUIDELINES

Criteria	Marks
• Links use outlined in (i) with concept of biotechnology	2
• Outlines meaning of biotechnology	1

Question 30 (b) (i)

Outcomes assessed: H11

MARKING GUIDELINES

Criteria	Marks
Appropriate procedure with relevant results including detailed information about procedures and/or results	5
- Appropriate procedure AND - Relevant results	3–4
- Limited procedure AND/OR - Limited result	1–2

Question 30 (b) (ii)

Outcomes assessed: H14

MARKING GUIDELINES

Criteria	Marks
Any relevant statement	1

Question 30 (c)

Outcomes assessed: H1, H14

MARKING GUIDELINES

Criteria	Marks
- Discusses benefits and/or limitations of at least TWO applications - Describes the impact of at least TWO applications - Outlines at least ONE rDNA method - Provides a response that demonstrates coherence and logical progression and includes correct use of scientific principles and ideas	7
- Describes the impact of at least TWO applications - Attempts to discuss benefits and/or limitations of at least ONE application - Outlines at least ONE rDNA method	5–6
- Describes the impact of applications - Outlines at least ONE rDNA method	3–4
- Outlines ONE method of rDNA technology OR - Outlines ONE application of rDNA technology	2
Identifies ONE method of rDNA technology OR ONE application of rDNA technology	1

Question 30 (d) (i)

Outcomes assessed: H12

MARKING GUIDELINES

Criteria	Marks
• Correct identification	1

Question 30 (d) (ii)

Outcomes assessed: H14

MARKING GUIDELINES

Criteria	Marks
• Correct outline of one piece of evidence	2
• Correct identification of one piece of evidence	1

Question 30 (d) (iii)

Outcomes assessed: H7

MARKING GUIDELINES

Criteria	Marks
• Assessment of the decision with reference to at least THREE pieces of evidence	4
• Assessment of the decision with reference to at least TWO pieces of evidence	3
• Assessment of the decision with reference to at least ONE piece of evidence OR • Outline of TWO pieces of evidence	2
• Outline of ONE piece of evidence	1

Question 31 (a) (i)

Outcomes assessed: H9

MARKING GUIDELINES

Criteria	Marks
• Outlines THREE differences	3
• Outlines TWO differences	2
• Outlines ONE difference	1

Question 31 (a) (ii)

Outcomes assessed: H9

MARKING GUIDELINES

Criteria	Marks
• Explains one reason	2
• Outlines one reason	1

Question 31 (b) (i)

Outcomes assessed: H11, H12, H14

MARKING GUIDELINES

Criteria	Marks
• Appropriate procedure with relevant findings including detailed information about procedures and/or findings	5
• Appropriate procedures and/or relevant findings	3–4
• Limited procedure And/or • Limited findings	1–2

Question 31 (b) (ii)

Outcomes assessed: H14

MARKING GUIDELINES

Criteria	Marks
• Any relevant statement	1

Question 31 (c)

Outcomes assessed: H1, H14

MARKING GUIDELINES

Criteria	Marks
- Well-linked discussion of at least TWO major advances, their role in structure, function and development along with our understanding of the processes involved - Provides a response that demonstrates coherence and logical progression and includes correct use of scientific principles and ideas	7
- Describes major advances and links these with structure, function or development of an organism - Attempts to discuss how advances have changed our understanding	5–6
Describes major advances and links these with structure, function or development of an organism	3–4
- Outlines one major advance OR - Outlines one way in which genes can direct the structure, function or development of an organism	2
- Identifies one way in which genes can direct the structure, function or development of an organism OR - Identifies one advance in our knowledge of genetics	1

Question 31 (d) (i)

Outcomes assessed: H12, H14

MARKING GUIDELINES

Criteria	Marks
Identifies correct genotype	1

Question 31 (d) (ii)

Outcomes assessed: H12

MARKING GUIDELINES

Criteria	Marks
Outlines the effect on milk production	2
Identifies the effect on milk production	1

Question 31 (d) (iii)

Outcomes assessed: H9, H14

MARKING GUIDELINES

Criteria	Marks
Clearly explains the link between polygenic inheritance on variability within a trait	4
- Describes ONE effect of polygenic inheritance and links this with variability within a trait OR - Explains ONE effect of polygenic inheritance	3
Describes ONE effect of polygenic inheritance	2
- Defines polygenic inheritance OR - Identifies ONE effect of polygenic inheritance	1

Question 32 (a) (i)

Outcomes assessed: H10

MARKING GUIDELINES

Criteria	Marks
Names TWO correct examples	2
Names ONE correct example	1

Question 32 (a) (ii)

Outcomes assessed: H10

MARKING GUIDELINES

Criteria	Marks
Links clear description of ONE cultural change named in part (a) (i) with human evolution	3
Describes ONE change named in part (a) (i)	2
Identifies ONE feature of human evolution	1

Question 32 (b) (i)

Outcomes assessed: H10, H11, H12, H14

MARKING GUIDELINES

Criteria	Marks
Appropriate procedure with relevant findings including detailed information about procedure and/or findings	5
- Appropriate procedure AND/OR - Relevant findings	3–4
- Limited procedure AND/OR - Limited findings	1–2

Question 32 (b) (ii)

Outcomes assessed: H10, H14

MARKING GUIDELINES

Criteria	Marks
Relevant correct statement	1

Question 32 (c)

Outcomes assessed: H1, H14

MARKING GUIDELINES

Criteria	Marks
- Well-linked discussion of at least TWO advances and how they have informed, changed or improved our understanding - Provides a response that demonstrates coherence and logical progression and includes correct use of scientific principles and ideas	7
- Describes at least TWO advances in technology and links them to changes in our understanding - Attempts to discuss how advances have changed our understanding	5–6
- Describes at least TWO advances in technology and links them to changes in our understanding OR - Describes how advances have changes our understanding	3–4
- Outlines ONE advance in technology OR - Outlines ONE change in understanding	2
- Identifies one advance in technology OR - Identifies one change in our understanding	1

Question 32 (d) (i)

Outcomes assessed: H10, H12

MARKING GUIDELINES

Criteria	Marks
Correct names ONE species of <i>Homo</i>	1

Question 32 (d) (ii)

Outcomes assessed: H10, H12

MARKING GUIDELINES

Criteria	Marks
Correctly outlines ONE piece of information	2
Correctly identifies of ONE piece of information	1

Question 32 (d) (iii)

Outcomes assessed: H14, H10

MARKING GUIDELINES

Criteria	Marks
- Assessment of the discovery - Clearly links the discovery of <i>Homo floresiensis</i> with EITHER the ‘Out of Africa’ model OR the ‘Theory of Regional Continuity’	4
Clearly links the discovery of <i>Homo floresiensis</i> with EITHER the ‘Out of Africa’ model OR the ‘Theory of Regional Continuity’	3
Outlines ONE piece of information and relates this to EITHER the ‘Out of Africa’ model OR the ‘Theory of Regional Continuity’	2
- Outline of the ‘Out of Africa’ model OR - Outline of the ‘Theory of Regional Continuity’	1

Question 33 (a) (i)

Outcomes assessed: H6

MARKING GUIDELINES

Criteria	Marks
States TWO correct reasons	2
States ONE correct reason	1

Question 33 (a) (ii)

Outcomes assessed: H1, H6

MARKING GUIDELINES

Criteria	Marks
Names ONE correct scientist, and outlines contribution	3
Names ONE correct scientist and limited outline	2
Names ONE correct scientist	1

Question 33 (b) (i)

Outcomes assessed: H11

MARKING GUIDELINES

Criteria	Marks
Appropriate procedure with relevant results, including detailed information about procedure and/or results	5
- Appropriate procedure AND - Relevant results	3–4
- Limited procedure AND/OR - Limited results	1–2

Question 33 (b) (ii)

Outcomes assessed: H12, H14

MARKING GUIDELINES

Criteria	Marks
Any relevant statement	1

Question 33 (c)

Outcomes assessed: H1, H14

MARKING GUIDELINES

Criteria	Marks
- Describes the development of both the light and electron microscope - Discusses the impact of both on our understanding of photosynthetic cell structures - Makes reference to the internal structure of chloroplasts - Provides a response that demonstrates coherence and logical progression and includes correct use of scientific principles and ideas	7
- Describes the development of both the light and electron microscope AND - Discusses the impact of both on our understanding of photosynthetic cell structures or makes reference to the internal structure of chloroplasts	5–6
- Outlines development of both light and electron microscopes with brief understanding of photosynthetic cell structures OR - Details the development of the light microscope or the electron microscope with reference to our understanding of photosynthetic cell structures	3–4
Outlines developments in microscopy	2
Identifies ONE advance in microscopy	1

Question 33 (d) (i)

Outcomes assessed: H6, H12

MARKING GUIDELINES

Criteria	Marks
Correctly identifies X as Phosphoglycerate	1

Question 33 (d) (ii)

Outcomes assessed: H6

MARKING GUIDELINES

Criteria	Marks
States significance of 'Y' and 'Z'	2
- States significance of 'Y' OR - States significance of 'Z'	1

Question 33 (d) (iii)

Outcomes assessed: H2, H14

MARKING GUIDELINES

Criteria	Marks
- Clearly links knowledge and understanding of the Calvin Cycle with specific advances in technology - Assessment of advances	4
Clearly links knowledge and understanding of the Calvin Cycle with specific advances in technology	3
- Identifies both ONE advance in technology and ONE improvement in knowledge about the Calvin Cycle OR - Describes either ONE advance in technology or ONE improvement in knowledge about the Calvin Cycle	2
- Identifies ONE advance in technology in relation to the Calvin Cycle OR - Identifies ONE improvement in knowledge about the Calvin Cycle	1